Rainfall Estimation

Introduction

Inverse-distance weighting (IDW) and related interpolation rules are widely used as simple baselines for rainfall-field estimation from microwave-link data. Their appeal is practical: they are fast, easy to implement, and often produce visually plausible maps. However, these methods do not explicitly search for a rainfall field whose forward-model-implied link attenuations match the observed link attenuations. This note investigates whether a nonlinear optimization model can improve on IDW and on the link-aware ILDW baseline when the goal is not only to predict rainfall, but also to remain faithful to the measurements that generated the estimate.

Methodology

Our benchmark is built patch by patch. A *patch* is a rectangular crop of an hourly rainfall accumulation map from the OPERA European weather-radar product, selected to capture a detected rain event while satisfying patch-width and patch-height constraints (in the present benchmark, between 50 km and 95 km in each dimension). For each patch in the benchmark list, we first extract the corresponding rainfall crop from the OPERA map. We then refine that crop from the native 2 km grid to a finer 125 m grid by subdividing each coarse OPERA pixel into smaller pixels that inherit its rainfall value, and smooth the refined field. This smoothed, refined rainfall map is the ground-truth field used both to synthesize link attenuations and to evaluate the solvers. The full microwave-link data set, obtained from the 4TU-NL link data, is then overlaid on each patch, and only links whose endpoints fall inside the patch are retained. Each retained link keeps the same underlying metadata as in the full data set, including endpoints, frequency, and polarization, but is used in the benchmark only when it lies within the spatial extent of the selected patch.

A link is treated geometrically as a line segment between two endpoints. Given the ground-truth rainfall field, we compute the total attenuation of each link with the ITU-R P.838-3 forward model for rain-induced specific attenuation. In principle this is a path integral along the link. In the benchmark, and likewise for the methods we compare against, that forward model is evaluated in discretized form on the refined 125 m grid: each link is traced through the grid cells it crosses, and the total attenuation is obtained by summing the attenuation contributions of those crossed cells. These synthetic attenuation values are then given to the inverse solvers. In this note we compare three such solvers: IDW, ILDW, and a nonlinear optimization model.

The motivation for the nonlinear model is twofold. First, we hypothesized that rainfall estimation posed as a nonlinear optimization problem could outperform interpolation-based baselines in regions that are weakly constrained by nearby links, that is, at pixels for which the distance to the third-closest link is large. Second, we expected the resulting rainfall field to explain the attenuation measurements more faithfully: when the same forward model is applied to the predicted rainfall field, the mismatch between predicted and observed link attenuations should be smaller than for IDW and ILDW. The first question is examined below through distance-binned pixelwise error summaries, while the second is connected to the attenuation-error and objective-based analyses developed later in the note. Finally, because the patch detector is designed to capture rainy events rather than dry background, the benchmark is rain-dominated: averaging over the 100 patch-level dry fractions gives a mean non-rainy proportion of about 10.50 %, where non-rainy pixels are those with very little or no rainfall.

IDW Baseline

Inverse-distance weighting (IDW) is one of the standard simple interpolation strategies used in commercial-microwave-link rainfall reconstruction. In the setting considered here, each link carries a total attenuation value together with its geometric path, frequency, and polarization. The underlying forward model is the ITU-R P.838-3 rain-attenuation relation along the link path. In continuous form this is a line integral of rain-specific attenuation along the segment. In the implementation used in this project, and likewise in the comparisons below, that integral is discretized on the refined 125 m grid by tracing each link through the crossed cells and summing the attenuation contribution of each crossed cell.

The IDW baseline first converts each observed link attenuation into a single rain-rate estimate. If L_ℓ denotes the link length, m_ℓ the link midpoint, $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to the link frequency f_ℓ, and pol_ℓ the polarization of the link, then the link-level rain estimate is

$$R_\ell \triangleq \left(\frac{A_\ell / L_\ell}{\kappa_\ell} \right)^{1/\alpha_\ell}.$$

These link-level estimates are then treated as point observations located at the link midpoints. For a pixel p with center c_p, midpoint IDW computes a weighted average of the nearby R_ℓ values, with weights proportional to $\|m_\ell - c_p\|_2^{-2}$ inside a prescribed search radius $r_{\max}$. If one or more midpoints fall directly inside the pixel, the estimate is taken to be the mean of those link-level values. This is the discretized midpoint-IDW baseline used throughout the note. TODO (Iván): decide whether this explanatory paragraph should remain or be removed.

Algorithm 1 Discretized Midpoint IDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \mathrm{pol}_\ell)$, radius $r_{\max}$

for each link ℓ **do**
    $L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$
    $(\kappa_\ell, \alpha_\ell) \leftarrow \mathrm{ITU838}(f_\ell, \mathrm{pol}_\ell)$
    $R_\ell \leftarrow \left(\frac{A_\ell/L_\ell}{\kappa_\ell}\right)^{1/\alpha_\ell}$
end for
for each pixel p with center c_p **do**
    if one or more link midpoints lie inside pixel p **then**
        $\hat{R}_p \leftarrow$ mean of R_ℓ of those links
    else
        $N_p \leftarrow \{\ell : \|m_\ell - c_p\|_2 \leq r_{\max}\}$
        if $N_p = \emptyset$ **then**
            $\hat{R}_p \leftarrow 0$
        else
            $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\|m_\ell - c_p\|_2^{-2}}{\sum_{j \in N_p} \|m_j - c_p\|_2^{-2}} R_\ell$
        end if
    end if
end for
Return field $\hat{R}$

ILDW Baseline

ILDW is the link-aware variant introduced in this project as a stronger interpolation baseline. It starts from the same link-level rain estimates R_ℓ obtained by inverting the discretized attenuation forward model link by link. The difference is geometric: instead of treating each link as if all of its information were concentrated at the midpoint, ILDW uses the full link segment when deciding which links should influence a pixel and how strongly they should do so.

More precisely, for a pixel center c_p, ILDW replaces the midpoint distance $\|m_\ell - c_p\|_2$ by the point-to-segment distance

$$\mathrm{dist}\big(c_p, [p_{\ell,1}, p_{\ell,2}]\big).$$

If one or more links cross the pixel, the estimate is taken to be the mean of the corresponding R_ℓ values. Otherwise, the estimate is a weighted average over nearby links within radius $r_{\max}$, with weights proportional to the inverse squared segment distance. We considered this baseline important because it preserves the same simple interpolation philosophy as IDW while respecting more of the measurement geometry of the links themselves.

Algorithm 2 Segment-Distance ILDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \mathrm{pol}_\ell)$, radius $r_{\max}$

for each link ℓ **do**
    $L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$
    $(\kappa_\ell, \alpha_\ell) \leftarrow \mathrm{ITU838}(f_\ell, \mathrm{pol}_\ell)$
    $R_\ell \leftarrow \left(\dfrac{A_\ell / L_\ell}{\kappa_\ell}\right)^{1/\alpha_\ell}$
end for
for each pixel p with center c_p **do**
    if one or more links cross pixel p **then**
        $\hat{R}_p \leftarrow$ mean of R_ℓ of those links
    else
        $N_p \leftarrow \{\ell : \mathrm{dist}(c_p, [p_{\ell,1}, p_{\ell,2}]) \le r_{\max}\}$
        if $N_p = \emptyset$ **then**
            $\hat{R}_p \leftarrow 0$
        else
            $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\mathrm{dist}(c_p, [p_{\ell,1}, p_{\ell,2}])^{-2}}{\sum_{j \in N_p} \mathrm{dist}(c_p, [p_{j,1}, p_{j,2}])^{-2}} R_\ell$
        end if
    end if
end for
Return field $\hat{R}$

Optimization Algorithm

The nonlinear optimizer is the method we introduced to address some of the limitations of IDW and ILDW. Both interpolation-based baselines first reduce each link to a link-level rain estimate and then spread that information spatially by geometric weighting. This is simple and often effective, but it does not explicitly search for a rainfall field whose forward-model-implied attenuations agree as well as possible with the observed link attenuations. Our working conjecture was that a nonlinear program could improve on these baselines, especially in regions that are farther from nearby links and therefore less directly constrained by local interpolation.

The optimizer used in this note is L-BFGS-B, a limited-memory quasi-Newton method for smooth constrained optimization. In the present setting, the optimization variable is the rainfall value at every pixel of the patch, so the problem dimension is the number of pixels in the patch. Rather than storing a full Hessian matrix, L-BFGS-B builds a low-memory approximation to local curvature information from the most recent iterates and gradients, and uses that approximation to propose a descent direction. A line search is then carried out along that direction, and the process is repeated until the stopping criteria on iteration count, function reduction, or projected gradient norm are met. The suffix "-B" indicates that the method handles box constraints. Here those constraints enforce nonnegativity of the rainfall field, so every pixel estimate is required to remain in $[0, \infty)$. The optimizer is initialized from the ILDW estimate.

Mathematical Formulation

For a patch P, let $x \in \mathbb{R}_{\geq 0}^{|\Omega(P)|}$ denote the unknown rainfall field written as a vector over pixels. The optimizer minimizes a weighted sum of four terms,

$$J(x) \triangleq \alpha_{\text{atten}} J_{\text{atten}}(x) \;+\; \alpha_{1d} J_{1d}(x) \;+\; \alpha_{2d} J_{2d}(x) \;+\; \alpha_{\text{total}} J_{\text{total}}(x).$$

Here J_{atten} penalizes disagreement between observed and forward-model-predicted link attenuations, while J_{1d}, J_{2d}, and J_{total} are regularization terms that encourage the estimated field to remain spatially and physically well behaved. In the normalized solver studied here, the weights are chosen instance by instance from the corresponding ILDW objective values, namely

$$\alpha_{\text{atten}} = \frac{1}{J_{\text{atten}}^{\text{ILDW}}}, \qquad \alpha_{1d} = \frac{1}{J_{1d}^{\text{ILDW}}}, \qquad \alpha_{2d} = \frac{1}{2\, J_{2d}^{\text{ILDW}}}, \qquad \alpha_{\text{total}} = \frac{1}{J_{\text{total}}^{\text{ILDW}}}.$$

More concretely, J_{atten} is the data-fidelity term:

$$J_{\text{atten}}(x) = \frac{1}{|\mathcal{L}(P)|} \sum_{\ell \in \mathcal{L}(P)} \frac{\left(\widehat{A}_\ell(x) - A_\ell^{\text{obs}}\right)^2}{L_\ell},$$

where $\widehat{A}_\ell(x)$ is the attenuation predicted by the forward model from the rainfall field x, A_ℓ^{obs} is the observed attenuation of link ℓ, and L_ℓ is the link length in km. Thus J_{atten} measures how far the forward-model-implied attenuations are from the observed attenuations, with a normalization by link length and by the number of links in the patch.

The term J_{1d} is a first-difference penalty:

$$J_{1d}(x) = \frac{1}{|\Omega(P)|} \sum_{(u,v) \in \mathcal{E}(P)} \left(x_u - x_v\right)^2,$$

where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P. This term sums squared differences between neighboring pixel values and therefore discourages abrupt pixel-to-pixel jumps in the estimated rainfall field.

The term J_{2d} is a second-difference penalty:

$$J_{2d}(x) = \frac{1}{|\Omega(P)|} \sum_{(p,q,r) \in \mathcal{T}(P)} \left((x_q - x_p) - (x_r - x_q)\right)^2,$$

where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P. It penalizes changes in the first difference, so it discourages unnecessary curvature or oscillation along rows and columns.

Finally, J_{total} is an ℓ_2 shrinkage term on the rainfall field itself:

$$J_{\text{total}}(x) = \frac{1}{|\Omega(P)|} \sum_{p \in \Omega(P)} x_p^2.$$

This term biases the solution toward smaller overall rainfall values, except where larger values are needed to match the observed link attenuations.

These four terms are added together only after weighting, so the optimizer balances attenuation fit against smoothness, curvature control, and shrinkage. The normalization above makes the different objective terms comparable in scale at the ILDW initialization and lets the optimizer refine that starting point by directly minimizing a measurement-aware objective instead of relying only on geometric interpolation. The cost-reduction plots in the qualitative case-study section show how this weighted objective decreases over successive L-BFGS-B iterations.

Results

We now turn to the empirical comparison of the three methods on the hundred-patch benchmark. The results are organized around three complementary questions. First, how does prediction quality vary with link proximity? To answer this, we group pixels into distance bins according to their distance to the third-closest link and summarize patch-level error statistics within each bin. We first examine rainy pixels through relative absolute error, and then examine non-rainy pixels through absolute error. Second, how well do the predicted rainfall fields explain the observed measurements? This is addressed through attenuation-error and objective-based summaries. Third, what do the three methods look like on individual rain events? This is illustrated through the qualitative case studies and the corresponding optimization traces shown later in the note.

Relative Absolute Error Rainy-Pixel Distance Profile

For a patch P, let $\Omega(P)$ denote its pixel set, let $R_P : \Omega(P) \to \mathbb{R}_{\geq 0}$ be the ground-truth rainfall field, and let $\widehat{R}_{P,s} : \Omega(P) \to \mathbb{R}_{\geq 0}$ be the rainfall field predicted by solver s. We define a pixel p to be rainy if $R_P(p) \geq 0.6$ mm/h, and we write

$$\Omega_{\text{rain}}(P) = \{p \in \Omega(P) : R_P(p) \geq 0.6\}.$$

For a rainy pixel p, the relative absolute error with respect to solver s is

$$\text{RAE}_{P,s}(p) \triangleq \frac{|R_P(p) - \widehat{R}_{P,s}(p)|}{R_P(p)}.$$

To study how prediction quality changes with link proximity, we group pixels by distance to the third-closest link. Let $\mathcal{L}(P)$ denote the set of links contained in patch P. For a pixel $p \in \Omega(P)$, let $c(p) \in \mathbb{R}^2$ denote the center of pixel p. For a link $\ell \in \mathcal{L}(P)$, viewed geometrically as a line segment, let

$$\text{dist}(p, \ell) \triangleq \text{dist}\big(c(p), \ell\big)$$

denote the Euclidean distance from the pixel center to that link segment. If the links in patch P are written as $\ell_1, \ldots, \ell_{|\mathcal{L}(P)|}$, we define

$$d_3(p) \triangleq \text{the third-order statistic of } \big\{\text{dist}(p, \ell_j)\big\}_{j=1}^{|\mathcal{L}(P)|},$$

that is, the distance from the center of pixel p to its third-closest link segment.

For each index i, let b_{i-1} and b_i denote the lower and upper distance cutoffs of the ith bin, and write $B_i = (b_{i-1}, b_i]$. A pixel p belongs to bin B_i exactly when $d_3(p) \in B_i$.

For each patch P, solver s, and distance bin B_i, we then collect the rainy-pixel RAE values in that bin and summarize them by the patch-level median

$$m_{s,i}^{\text{rain}}(P) \triangleq \text{median}\big\{\text{RAE}_{P,s}(p) : p \in \Omega_{\text{rain}}(P),\ d_3(p) \in B_i\big\}\ .$$

In Figure 1, these patch-level medians are compared across the full benchmark collection $\mathcal{P}$ of 100 patches. There, the displayed distance bins are

$$[0, 125] \text{ m},\ (125, 375] \text{ m},\ (375, 750] \text{ m},\ (750, 1500] \text{ m},\ (1500, 3125] \text{ m},\ (3125, 6000] \text{ m},$$

with the sparsely populated tail beyond 6000 m merged into the last displayed bin. For fixed solver s and distance bin B_i, let

$$\mu_{s,i}^{\text{rain}} \triangleq \{m_{s,i}^{\text{rain}}(P)\}_{P \in \mathcal{P}}$$

denote the sequence of patch-level rainy-pixel median RAE values across patches. Each box-and-whisker then summarizes this sequence: the lower whisker tip is the minimum of $\mu_{s,i}^{\text{rain}}$, the lower edge of the box is its 25th percentile, the horizontal line segment with the filled circle at its center marks the median, the upper edge of the box is its 75th percentile, and the upper whisker tip is its maximum.

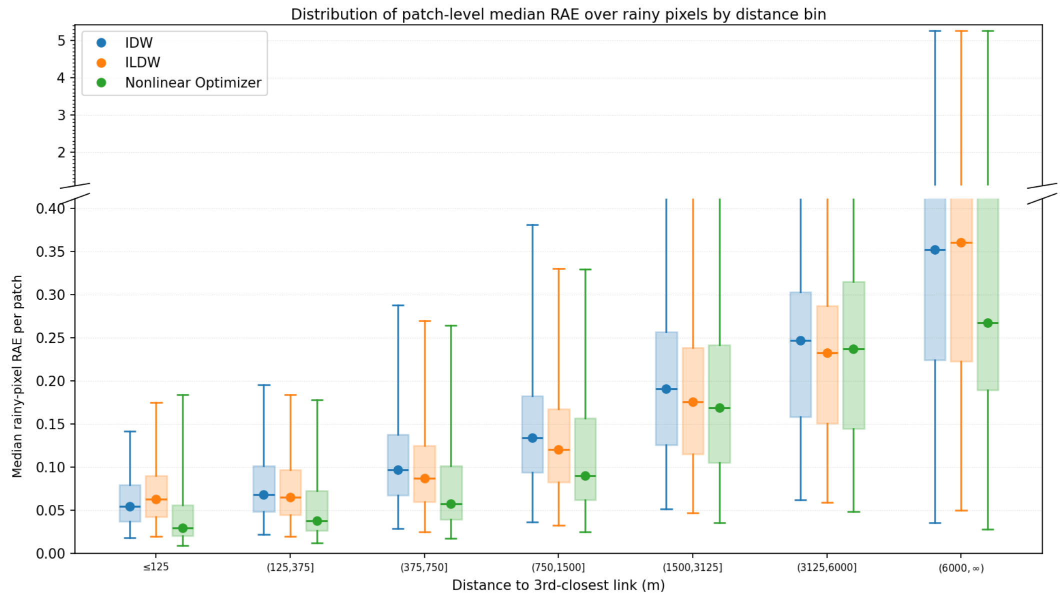


Figure 1: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6 mm/h), and the relative absolute error for a rainy pixel p by solver s is defined as $|R_P(p) - \widehat{R}_{P,s}(p)|/R_P(p)$. The corresponding rainy-pixel counts per bin are reported in Table 1.

Absolute Error Non-Rainy-Pixel Distance Profile

For the dry-pixel analysis, we focus on pixels whose ground-truth rainfall falls below the wet threshold. We therefore define the non-rainy pixel set by

$$\Omega_{\text{nonrain}}(P) = \{p \in \Omega(P) : R_P(p) < 0.6\}.$$

For these pixels, relative normalization by $R_P(p)$ is not appropriate because the denominator may be zero or very small. We therefore measure prediction quality by absolute error rather than relative absolute error:

$$\text{AE}_{P,s}(p) = |R_P(p) - \widehat{R}_{P,s}(p)|.$$

Using the same definition of the third-closest-link distance $d_3(p)$ introduced above, and the same distance bins B_i as in the rainy-pixel analysis, we compute for each patch P, solver s, and distance

Distance to third closest link (m)	Wet mean count (% of total)	SD
$[0, 125]$	4267.0 (1.79%)	960.4
$(125, 375]$	16737.7 (7.01%)	3803.2
$(375, 750]$	26809.9 (11.22%)	6707.0
$(750, 1500]$	51988.5 (21.76%)	14 468.0
$(1500, 3125]$	88839.4 (37.18%)	27 531.1
$(3125, 6000]$	45929.4 (19.22%)	15 679.4
$(6000, 9000]$	3835.8 (1.61%)	2725.7
$(9000, \infty)$	514.4 (0.22%)	966.9
Total	238922.1 (100%)	

Table 1: Rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 1. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 1 because it is sparsely populated.

bin B_i

$$m_{s,i}^{\text{nonrain}}(P) \triangleq \text{median}\big\{\text{AE}_{P,s}(p) : p \in \Omega_{\text{nonrain}}(P),\ d_3(p) \in B_i\big\},$$

and, for fixed s and i, the corresponding box-and-whisker plot summarizes the sequence

$$\mu_{s,i}^{\text{nonrain}} \triangleq \{m_{s,i}^{\text{nonrain}}(P)\}_{P \in \mathcal{P}}$$

of patch-level non-rainy-pixel median AE values across patches. Thus the rainy and non-rainy figures use the same distance-bin construction, but they differ both in the pixel set being summarized and in the error metric being aggregated.

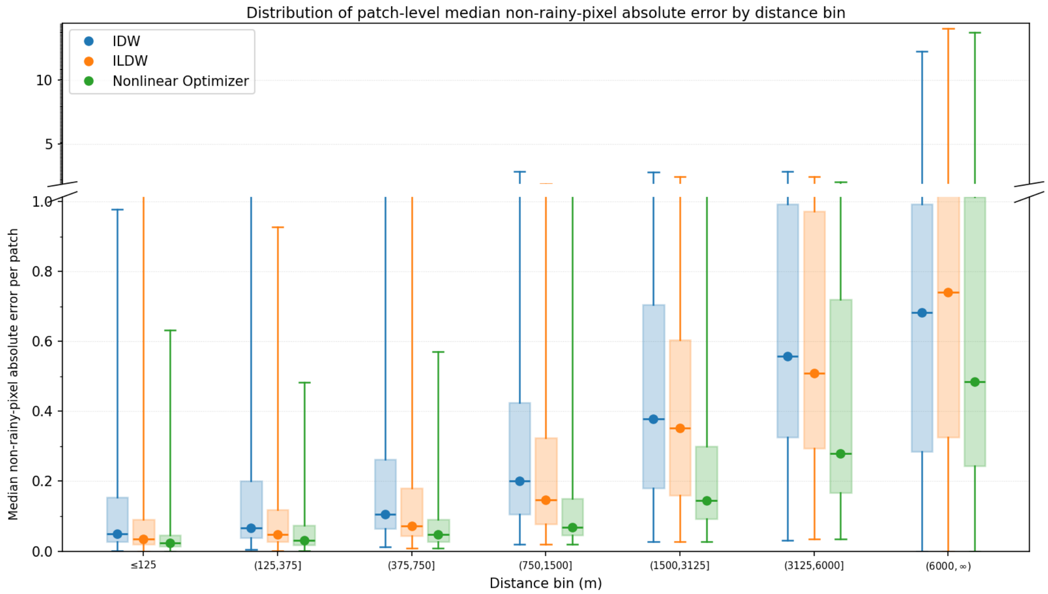


Figure 2: Non-rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median non-rainy-pixel absolute error value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a non-rainy pixel is a pixel whose ground-truth rainfall is below the configured wet threshold (set to 0.6 mm/h), and the absolute error for a non-rainy pixel p by solver s is defined as $|R_P(p) - \widehat{R}_{P,s}(p)|$. The corresponding non-rainy-pixel counts per bin are reported in Table 2.

Distance to third closest link (m)	Dry mean count (% of total)	SD
$[0, 125]$	563.6 (1.84%)	777.6
$(125, 375]$	2261.1 (7.40%)	3093.5
$(375, 750]$	3389.1 (11.09%)	4709.5
$(750, 1500]$	6446.5 (21.09%)	9257.7
$(1500, 3125]$	10776.2 (35.25%)	15 553.7
$(3125, 6000]$	6197.2 (20.27%)	8990.0
$(6000, 9000]$	849.9 (2.78%)	1289.7
$(9000, \infty)$	85.5 (0.28%)	250.9
Total	30569.2 (100%)	

Table 2: Non-rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 2. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the non-rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 2 because it is sparsely populated.

Largest-Patch Link Geometry

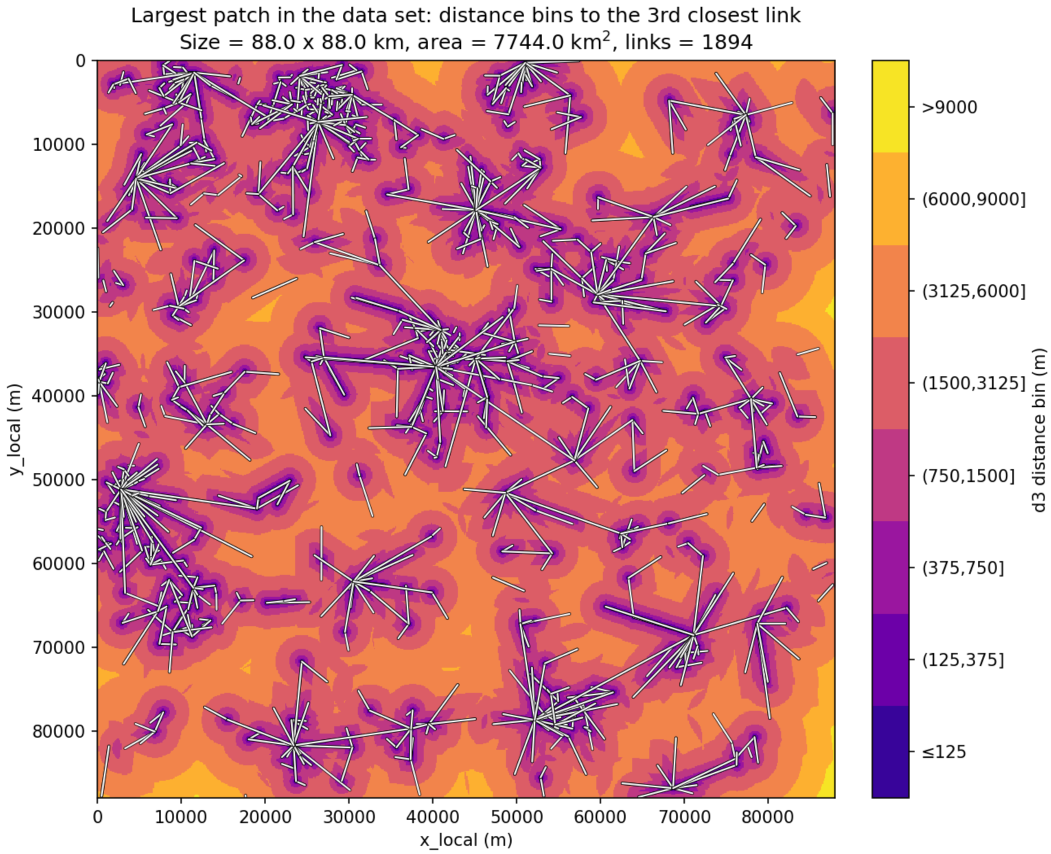


Figure 3: Distance-bin map for the largest evaluated patch in the data set. Each pixel is colored according to the bin induced by its distance to the third closest link, and the complete link geometry is overlaid in white.

The same largest patch is also useful for describing the structure of the underlying link network. In the summaries below, link length refers to the path length of a link, carrier frequency is the microwave-link frequency in GHz, and endpoint degree is the number of links incident to a given endpoint node. Here, an endpoint node is a location at which one or more links meet.

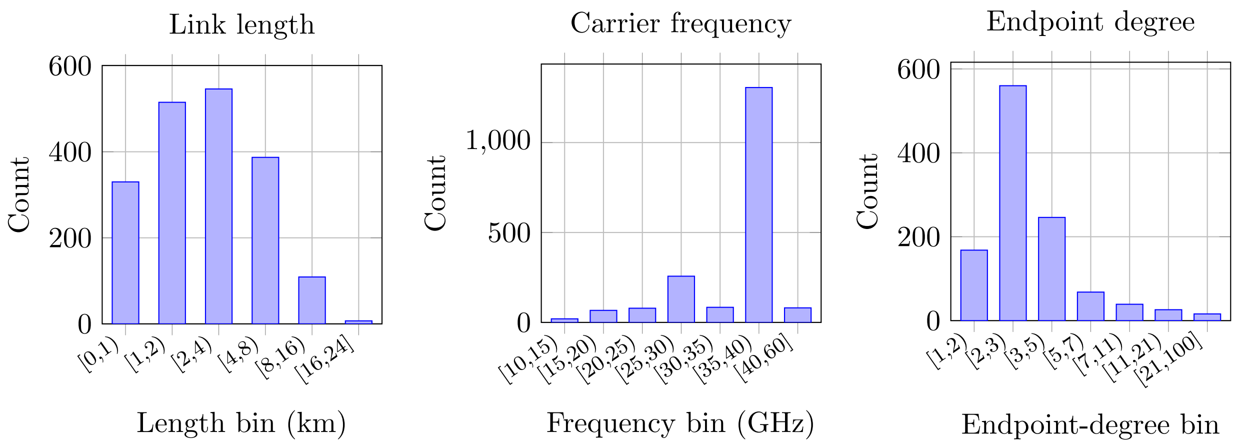


Figure 4: Histogram summaries for the largest patch. Left: link lengths. Middle: carrier frequencies. Right: endpoint-degree histogram. The x-axis gives endpoint degree and the y-axis gives the number of endpoint nodes in each degree bin.

Length bin (km)	Count	%
$[0,1)$	330	17.4
$[1,2)$	515	27.2
$[2,4)$	546	28.8
$[4,8)$	387	20.4
$[8,16)$	109	5.8
$[16,24]$	7	0.4

Frequency bin (GHz)	Count	%
$[10,15)$	20	1.1
$[15,20)$	67	3.5
$[20,25)$	79	4.2
$[25,30)$	257	13.6
$[30,35)$	84	4.4
$[35,40)$	1306	69.0
$[40,60]$	81	4.3

Endpoint-degree bin	Count	%
$[1,2)$	168	15.0
$[2,3)$	560	49.9
$[3,5)$	246	21.9
$[5,7)$	68	6.1
$[7,11)$	39	3.5
$[11,21)$	26	2.3
$[21,100]$	16	1.4

Table 3: Tabulated distributions for the largest-patch link network. The first panel counts links by path-length interval, the second counts links by carrier-frequency interval, and the third counts unique endpoint nodes by their graph degree.

Confusion Matrix

IDW

GT	Prediction Wet	Prediction Dry
Wet	0.893 ± 0.013	0.002 ± 0.000
Dry	0.040 ± 0.004	0.065 ± 0.010

ILDW

GT	Prediction Wet	Prediction Dry
Wet	0.892 ± 0.013	0.003 ± 0.000
Dry	0.037 ± 0.004	0.068 ± 0.010

Nonlinear Optimizer

GT	Prediction Wet	Prediction Dry
Wet	0.890 ± 0.013	0.005 ± 0.001
Dry	0.030 ± 0.004	0.075 ± 0.010

Table 4: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM).

Patch Geometry Overview

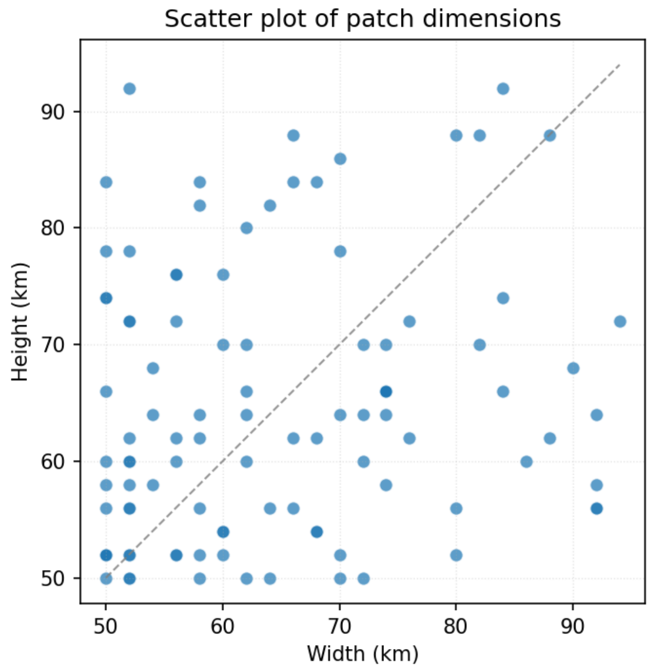


Figure 5: Patch width versus patch height for the 100 patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125m).

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 5: Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

J_{atten} Summary

Solver	Mean J_{atten} (dB^2/km)	SD
IDW	0.01522	0.03808
ILDW	0.00289	0.00757
Nonlinear Optimizer	0.00020	0.00103

Table 6: Patch-level summary of J_{atten} across the 100 evaluated patches. For a given patch, J_{atten} is the attenuation-fit term $J_{\text{atten}} = \frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{\left(\hat{A}_\ell - A_\ell^{\text{obs}}\right)^2}{|\ell|}$, where $\hat{A}_\ell$ is the attenuation implied by the estimated rainfall field on link ℓ, A_ℓ^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. The table reports the mean and standard deviation of this per-patch quantity for each solver. Lower values indicate better agreement with the observed link attenuations.

Absolute Attenuation Error per km

Solver	Mean (dB/km)	SD
IDW	0.05034	0.03413
ILDW	0.01356	0.00948
Nonlinear Optimizer	0.00230	0.00437

Table 7: Patch-level summary of the mean absolute attenuation error per km. For a given patch, we compute $\frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{\left|\hat{A}_\ell - A_\ell^{\text{obs}}\right|}{|\ell|}$, where $\hat{A}_\ell$ is the attenuation induced by the estimated rainfall field on link ℓ, A_ℓ^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. Thus, the absolute attenuation error of every link is normalized by the link's length. The table reports the mean and standard deviation of this quantity across the 100 evaluated patches for each solver. Lower values indicate better agreement with the observed link attenuations.

Qualitative Case Studies

The three case studies below pair a geographic view of the selected patch with the corresponding rainy-pixel solver outputs. The timestamp in each caption identifies the radar scene, and patch 0 denotes the selected patch within that scene. This layout is intended to keep the geographic context and the solver behavior visually tied to one another.

For the nonlinear optimizer, the estimated rainfall field is computed with the L-BFGS-B algorithm, initialized from the ILDW field and constrained to remain nonnegative at every pixel. The objective minimized by the optimizer is a weighted sum of four terms,

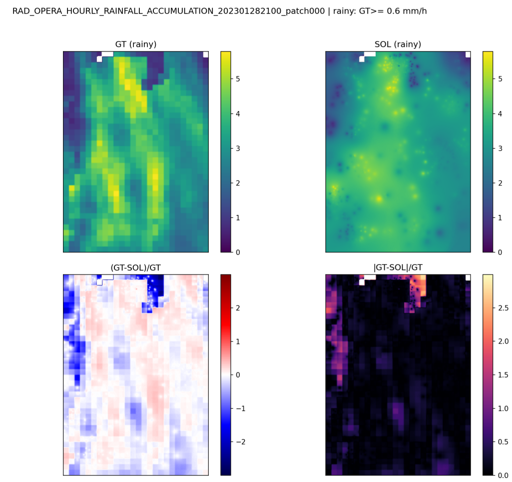
$$\alpha_{\text{atten}} J_{\text{atten}} + \alpha_{1d} J_{1d} + \alpha_{2d} J_{2d} + \alpha_{\text{total}} J_{\text{total}},$$

where the instance-specific weights are chosen from the corresponding ILDW objective values, namely

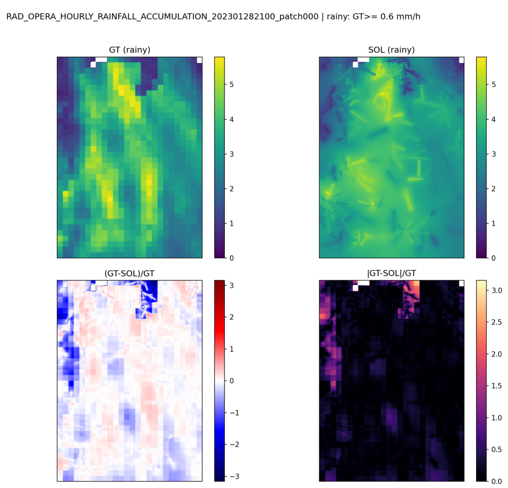
$$\alpha_{\text{atten}} = \frac{1}{J_{\text{atten}}^{\text{ILDW}}}, \qquad \alpha_{1d} = \frac{1}{J_{1d}^{\text{ILDW}}}, \qquad \alpha_{2d} = \frac{1}{2\, J_{2d}^{\text{ILDW}}}, \qquad \alpha_{\text{total}} = \frac{1}{J_{\text{total}}^{\text{ILDW}}}.$$

The optimizer-trace plots included below show, for each selected case study, how the weighted objective and its weighted constituent terms decrease over the optimization iterations.

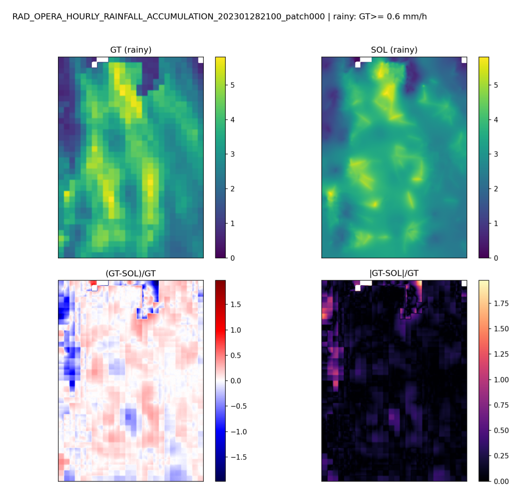
(a) Map of the area of the rain event



(b) IDW



(c) ILDW



(d) Nonlinear optimizer

Figure 6: Case study for a rain event registered on January 28th, 2023. The map on the left shows the selected patch footprint in geographic context. The three rainy-pixel solver panels correspond to that same patch and show, for each solver, the ground truth field R_P, the estimate $\widehat{R}_{P,s}$, the signed relative error $(R_P - \widehat{R}_{P,s})/R_P$, and the absolute relative error $|R_P - \widehat{R}_{P,s}|/R_P$. This layout keeps the location of the event visually tied to the reconstruction quality of each solver.

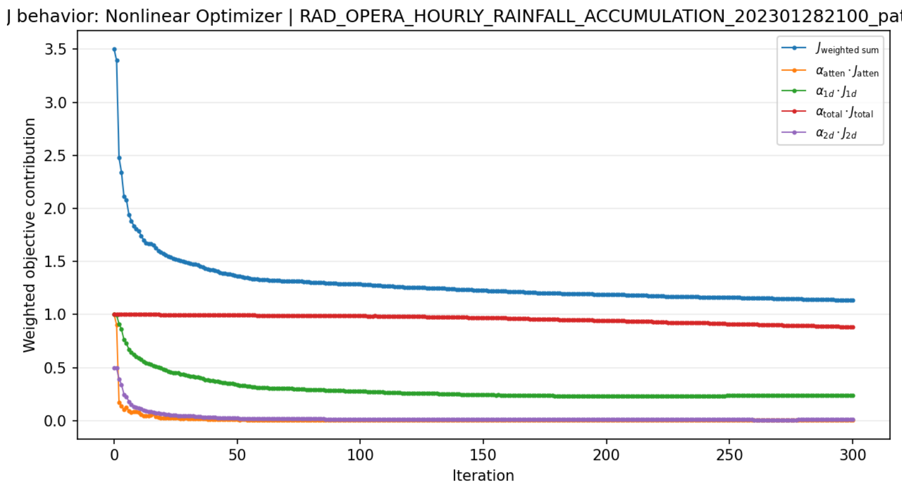
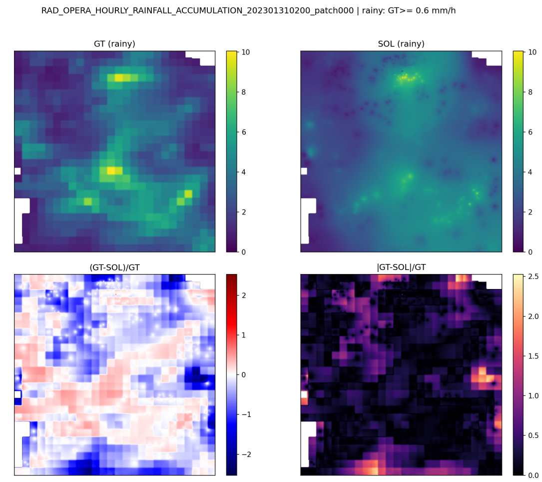
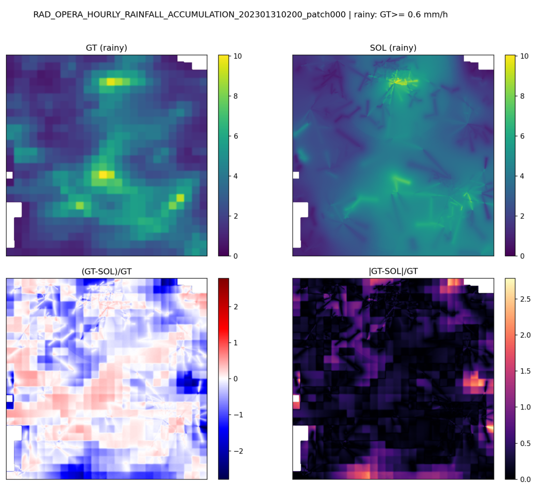


Figure 7: Optimizer trace for the January 28th, 2023 case study. The plot shows the reduction of the weighted objective $J_{\text{weighted sum}}$ and of the weighted constituent terms over L-BFGS-B iterations.

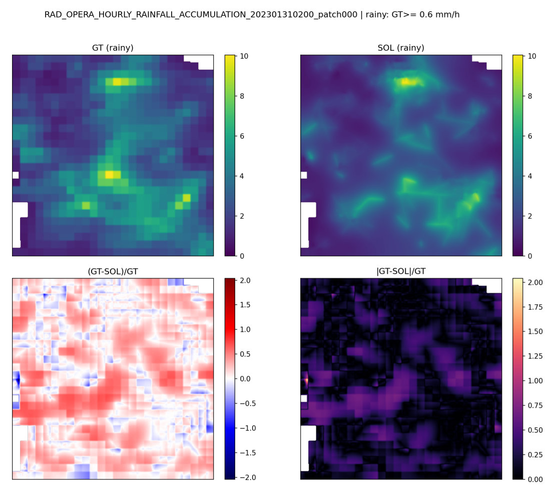
(a) Map of the area of the rain event



(b) IDW



(c) ILDW



(d) Nonlinear optimizer

Figure 8: Case study for a rain event registered on January 31st, 2023. The left panel provides the map view of the selected patch, while the three solver panels show, for IDW, ILDW, and the nonlinear optimizer, the rainy-pixel ground truth field R_P, the estimate $\widehat{R}_{P,s}$, the signed relative error $(R_P - \widehat{R}_{P,s})/R_P$, and the absolute relative error $|R_P - \widehat{R}_{P,s}|/R_P$. The common layout makes it easier to compare smoothing, displacement, and error localization across methods for the same event.

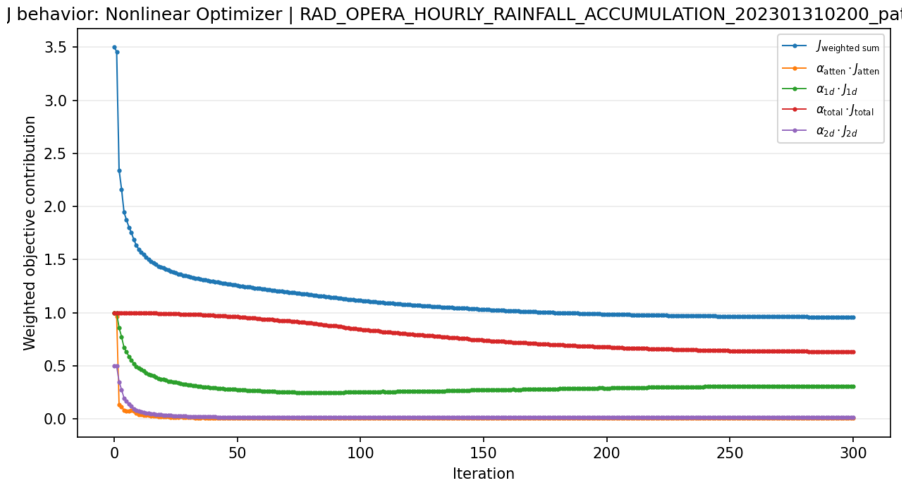
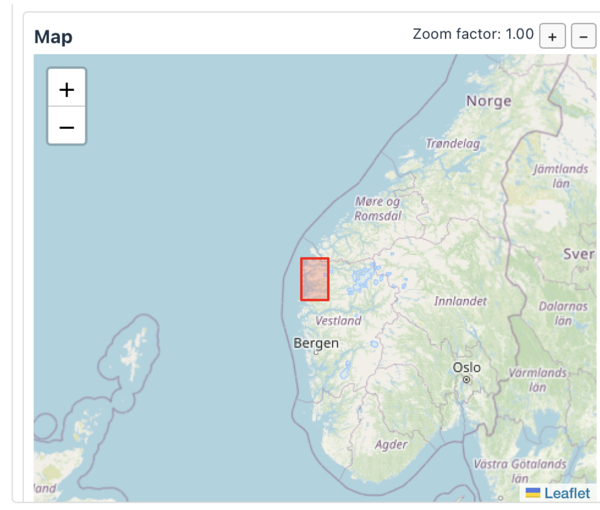
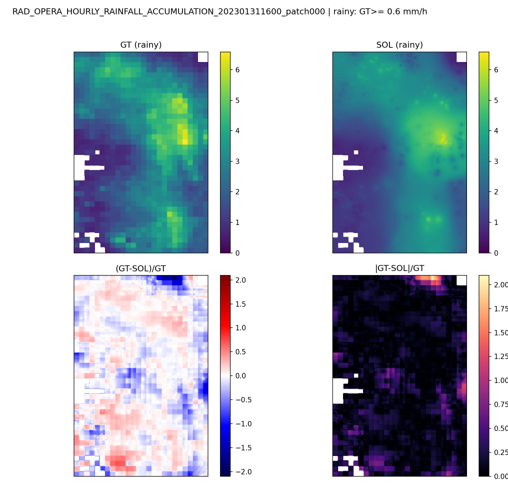


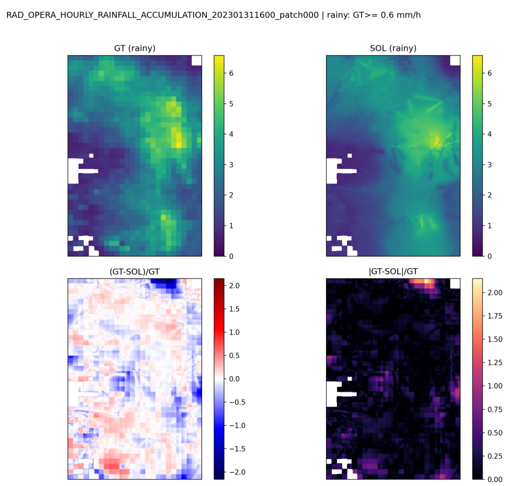
Figure 9: Optimizer trace for the January 31st, 2023 case study at 02:00. As above, the plot reports the iteration-by-iteration decrease of the weighted objective and its weighted constituent terms.



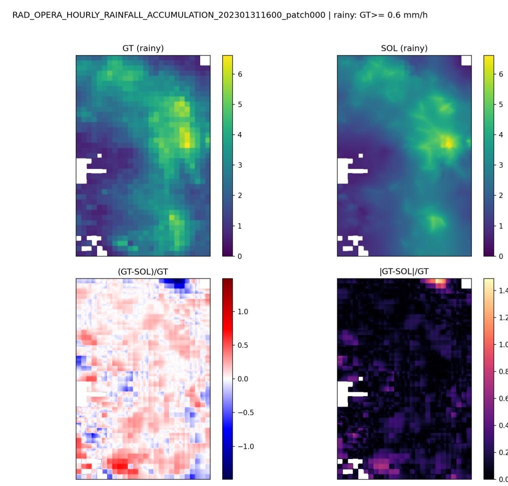
(a) Map of the area of the rain event



(b) IDW



(c) ILDW



(d) Nonlinear optimizer

Figure 10: Case study for a rain event registered on January 31st, 2023. The map on the left identifies the patch location and footprint, and the three solver panels show the same rainy-pixel diagnostic decomposition as above for IDW, ILDW, and the nonlinear optimizer: R_P, $\widehat{R}_{P,s}$, $(R_P - \widehat{R}_{P,s})/R_P$, and $|R_P - \widehat{R}_{P,s}|/R_P$. Together with the previous two figures, this case gives a compact qualitative comparison of solver behavior across distinct rainfall scenes.

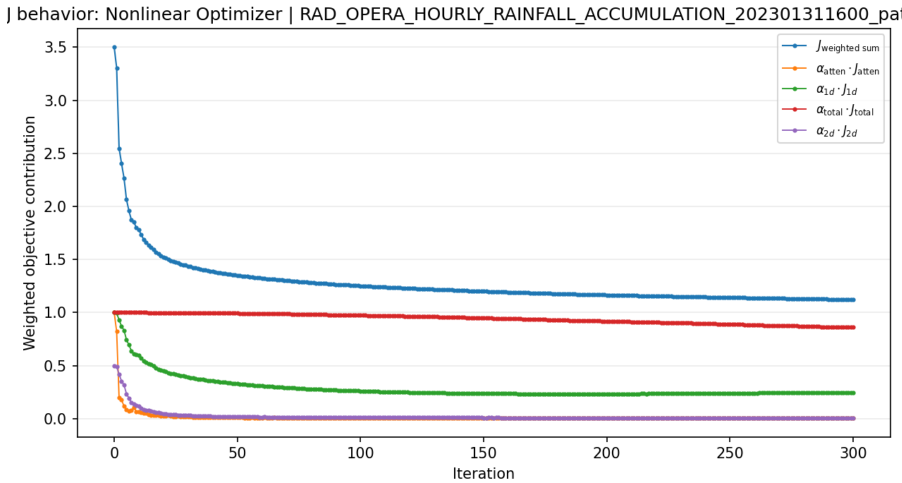


Figure 11: Optimizer trace for the January 31st, 2023 case study at 16:00. The weighted objective and weighted component terms are shown over the L-BFGS-B iterations for the same patch displayed in the spatial case-study panels above.

References